PROJECT BASED EXPERIMENTAL LEARNING

COURSE- DATABASE MANAGEMENT SYSTEM (NoSQL / MongoDB)

PROJECT

TOPIC- MONGODB STUDENT DATABASE OPERATIONS

IBM

SUBMITTED TO- MR. Harendra Singh Rajpoot

Submitted by- Vipul Pandey

BCADS27

Roll no. - 1250258507

3rd semester

Babu Banarasi Das University

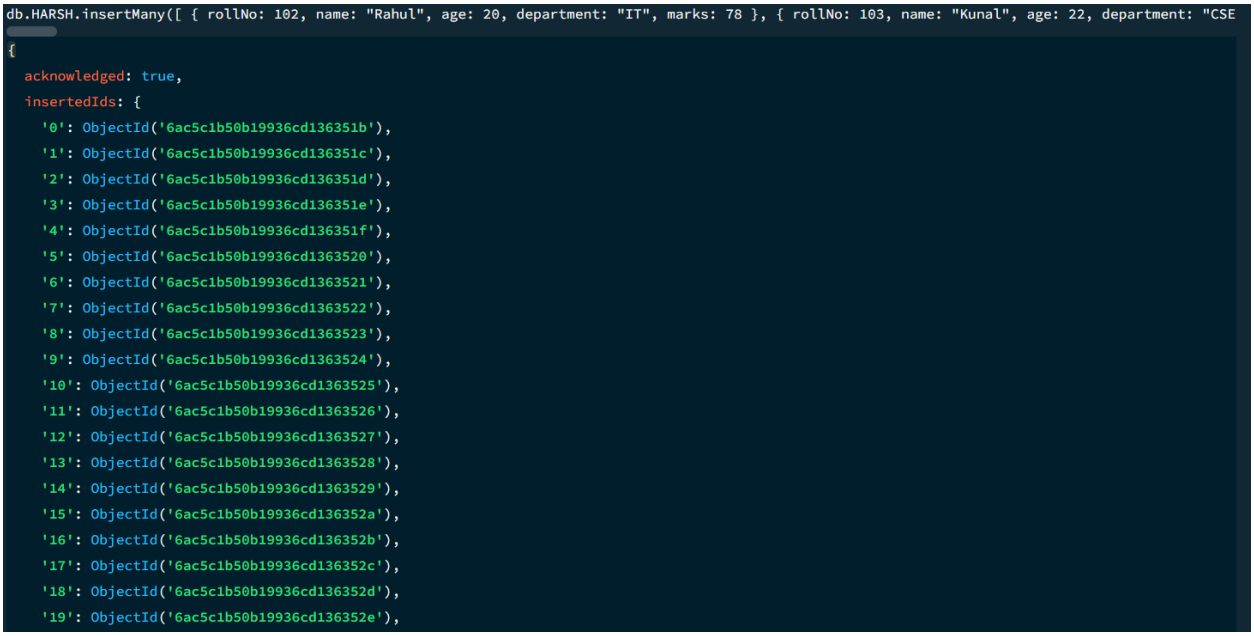
Lucknow, U.P

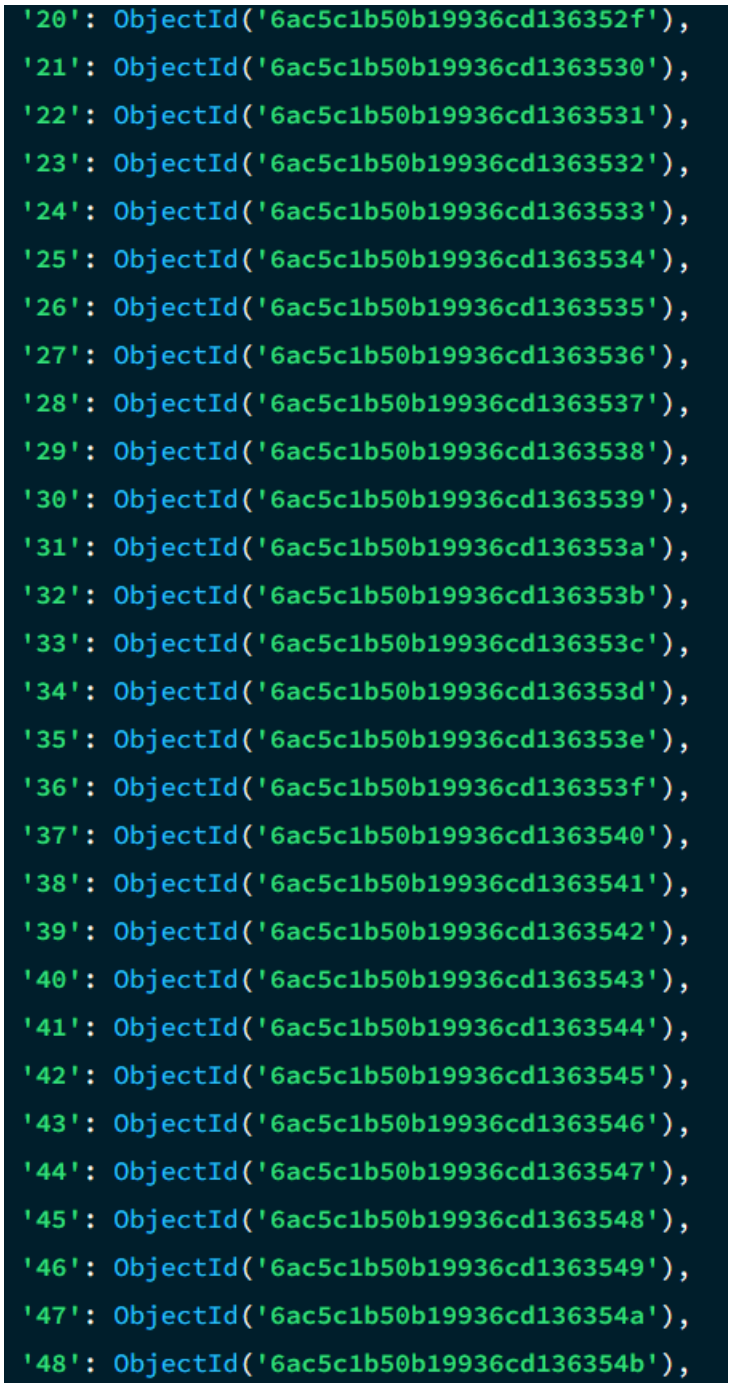
1.db.HARSH.insertOne({ rollNo: 101, name: "VIPUL", age: 22, department: "CSE", marks: 87 })

```
> use("PANDEY")
< switched to db PANDEY
> db["HARSH"].find()
<
>   db.HARSH.insertOne({ rollNo: 101, name: "VIPUL", age: 22, department: "CSE", marks: 87 })
< {
    acknowledged: true,
    insertedId: ObjectId('6ac5c0fb0b19936cd136351a')
  }
```

2.db.HARSH.insertMany([{ rollNo: 102, name: "Rahul", age: 20, department: "IT", marks: 78 }, { rollNo: 103, name: "Kunal", age: 22, department: "CSE", marks: 92 }, { rollNo: 104, name: "Meera", age: 21, department: "ECE", marks: 81 }, { rollNo: 105, name: "Varun", age: 23, department: "ME", marks: 74 }, { rollNo: 106, name: "Piyush", age: 20, department: "CSE", marks: 88 }, { rollNo: 107, name: "Rakesh", age: 21, department: "IT", marks: 69 }, { rollNo: 108, name: "Nandini", age: 22, department: "ECE", marks: 95 }, { rollNo: 109, name: "Aman", age: 21, department: "CSE", marks: 83 }, { rollNo: 110, name: "Kriti", age: 20, department: "IT", marks: 76 }, { rollNo: 111, name: "Saurabh", age: 23, department: "ME", marks: 72 }, { rollNo: 112, name: "Ananya", age: 21, department: "CSE", marks: 89 }, { rollNo: 113, name: "Rishabh", age: 22, department: "ECE", marks: 67 }, { rollNo: 114, name: "Tanvi", age: 20, department: "IT", marks: 91 }, { rollNo: 115, name: "Vivek", age: 21, department: "CSE", marks: 79 }, { rollNo: 116, name: "Shalini", age: 22, department: "ECE", marks: 84 }, { rollNo: 117, name: "Rajat", age: 23, department: "ME", marks: 73 }, { rollNo: 118, name: "Ishita", age: 21, department: "CSE", marks: 96 }, { rollNo: 119, name: "Nitin", age: 20, department: "IT", marks: 68 }, { rollNo: 120, name: "Sakshi", age: 22, department: "ECE", marks: 87 }, { rollNo: 121, name: "Anurag", age: 21, department: "CSE", marks: 75 }, { rollNo: 122, name: "Ritu", age: 20, department: "IT", marks: 90 }, { rollNo: 123, name: "Varun", age: 23, department: "ME", marks: 71 }, { rollNo: 124, name: "Neelam", age: 21, department: "CSE", marks: 86 }, { rollNo: 125, name: "Manav", age: 22, department: "ECE", marks: 80 }, { rollNo: 126, name: "Ayesha", age: 20, department: "IT", marks: 94 }, { rollNo: 127, name: "Rohan", age: 21, department: "CSE", marks: 77 }, { rollNo: 128, name: "Poonam", age: 22, department: "ECE", marks: 82 }, { rollNo: 129, name: "Sachin", age: 23, department: "ME", marks: 65 }, { rollNo: 130, name: "Kavita", age: 21, department: "CSE", marks: 93 }, { rollNo: 131, name: "Harish", age: 20, department: "IT", marks: 70 }, { rollNo: 132, name: "Radhika", age: 22, department: "ECE", marks: 89 }, { rollNo: 133, name: "Deep", age: 21, department: "CSE", marks: 81 }, { rollNo: 134, name: "Shweta", age: 20, department: "IT", marks: 85 }, { rollNo: 135, name: "Yogesh", age: 23, department: "ME", marks: 78 }, { rollNo: 136, name:

"Mansi", age: 21, department: "CSE", marks: 97 }, { rollNo: 137, name: "Ashish", age: 22, department: "ECE", marks: 74 }, { rollNo: 138, name: "Pallak", age: 20, department: "IT", marks: 88 }, { rollNo: 139, name: "Tarun", age: 21, department: "CSE", marks: 66 }, { rollNo: 140, name: "Renu", age: 22, department: "ECE", marks: 92 }, { rollNo: 141, name: "Vishal", age: 23, department: "ME", marks: 69 }, { rollNo: 142, name: "Shreya", age: 21, department: "CSE", marks: 84 }, { rollNo: 143, name: "Aakash", age: 20, department: "IT", marks: 79 }, { rollNo: 144, name: "Payal", age: 22, department: "ECE", marks: 91 }, { rollNo: 145, name: "Mohit", age: 21, department: "CSE", marks: 87 }, { rollNo: 146, name: "Komal", age: 20, department: "IT", marks: 73 }, { rollNo: 147, name: "Riya", age: 22, department: "ECE", marks: 86 }, { rollNo: 148, name: "Prakash", age: 23, department: "ME", marks: 68 }, { rollNo: 149, name: "Nisha", age: 21, department: "CSE", marks: 90 }, { rollNo: 150, name: "Arvind", age: 22, department: "IT", marks: 76 }, { rollNo: 151, name: "Simran", age: 20, department: "ECE", marks: 83 }])





3.db.HARSH.updateOne({ rollNo: 101 }, { $set: { marks: 90 } })

```
db.HARSH.updateOne( { rollNo: 101 }, { $set: { marks: 90 } }
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

4.db.HARSH.updateMany({ department: "CSE" }, { $set: { status: "Active" } })

```
db.HARSH.updateMany( { department: "CSE" }, { $set: { status: "Active" } } )
{
 acknowledged: true,
 insertedId: null,
 matchedCount: 17,
 modifiedCount: 17,
 upsertedCount: 0
```

5.db.HARSH.deleteOne({ rollNo: 151 })

```
db.HARSH.deleteOne({ rollNo: 151 })
{
  acknowledged: true,
  deletedCount: 1
}
```

6.db.HARSH.deleteMany({ marks: { $lt: 40 } })

```
db.HARSH.deleteMany({ marks: { $lt: 40 } })
{
  acknowledged: true,
  deletedCount: 0
}
```

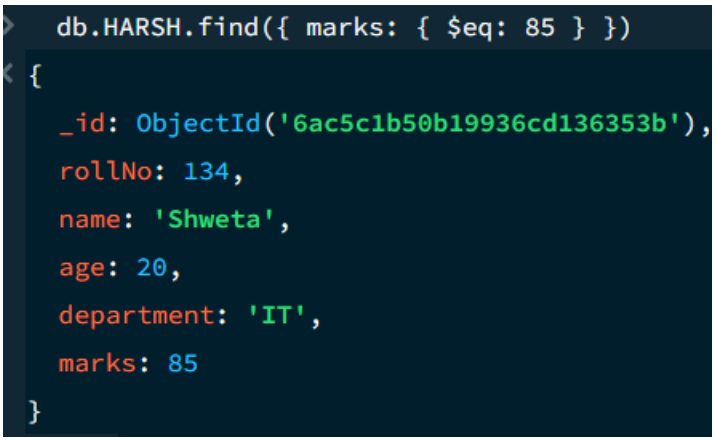
7.db.HARSH.find({ marks: { $gt: 80 } })



8.db.HARSH.find({ marks: { $lt: 80 } })

```
db.HARSH.find({ marks: { $lt: 80 } })
{
  _id: ObjectId('6ac5c1b50b19936cd136351b'),
  rollNo: 102,
  name: 'Rahul',
  age: 20,
  department: 'IT',
  marks: 78
}
{
  _id: ObjectId('6ac5c1b50b19936cd136351e'),
  rollNo: 105,
  name: 'Varun',
  age: 23,
  department: 'ME',
  marks: 74
}
{
  _id: ObjectId('6ac5c1b50b19936cd1363520'),
  rollNo: 107,
  name: 'Rakesh',
  age: 21,
  department: 'IT',
  marks: 69
}
```

9.db.HARSH.find({ marks: { $eq: 85 } })



10.db.HARSH.find({ marks: { $gte: 80 } })

```
> db.HARSH.find({ marks: { $gte: 80 } })
< {
    _id: ObjectId('6ac5c0fb0b19936cd136351a'),
    rollNo: 101,
    name: 'VIPUL',
    age: 22,
    department: 'CSE',
    marks: 90,
    status: 'Active'
  }
  {
    _id: ObjectId('6ac5c1b50b19936cd136351c'),
    rollNo: 103,
    name: 'Kunal',
    age: 22,
    department: 'CSE',
    marks: 92,
    status: 'Active'
  }
  {
    _id: ObjectId('6ac5c1b50b19936cd136351d'),
    rollNo: 104,
    name: 'Meera',
    age: 21,
    department: 'ECE',
    marks: 81
  }
```

11.db.HARSH.find({ marks: { $ne: 85 } })

```
db.HARSH.find({ marks: { $ne: 85 } })
{
  _id: ObjectId('6ac5c0fb0b19936cd136351a'),
  rollNo: 101,
  name: 'VIPUL',
  age: 22,
  department: 'CSE',
  marks: 90,
  status: 'Active'
}
{
  _id: ObjectId('6ac5c1b50b19936cd136351b'),
  rollNo: 102,
  name: 'Rahul',
  age: 20,
  department: 'IT',
  marks: 78
}
{
  _id: ObjectId('6ac5c1b50b19936cd136351c'),
  rollNo: 103,
  name: 'Kunal',
  age: 22,
  department: 'CSE',
  marks: 92,
  status: 'Active'
}
```

12.db.HARSH.find({ $and: [{ marks: { $gt: 70 } }, { age: { $lt: 23 } }] })

```
> db.HARSH.find({ $and: [ { marks: { $gt: 70 } }, { age: { $lt: 23 } } ] })
{
  _id: ObjectId('6ac5c0fb0b19936cd136351a'),
  rollNo: 101,
  name: 'VIPUL',
  age: 22,
  department: 'CSE',
  marks: 90,
  status: 'Active'
}
{
  _id: ObjectId('6ac5c1b50b19936cd136351b'),
  rollNo: 102,
  name: 'Rahul',
  age: 20,
  department: 'IT',
  marks: 78
}
{
  _id: ObjectId('6ac5c1b50b19936cd136351c'),
  rollNo: 103,
  name: 'Kunal',
  age: 22,
  department: 'CSE',
  marks: 92,
  status: 'Active'
}
```

13.db.HARSH.find({ $or: [{ department: "CSE" }, { department: "ECE" }] })

```
db.HARSH.find({ $or: [ { department: "CSE" }, { department: "ECE" } ] })
{
  _id: ObjectId('6ac5c0fb0b19936cd136351a'),
  rollNo: 101,
  name: 'VIPUL',
  age: 22,
  department: 'CSE',
  marks: 90,
  status: 'Active'
}
{
  _id: ObjectId('6ac5c1b50b19936cd136351c'),
  rollNo: 103,
  name: 'Kunal',
  age: 22,
  department: 'CSE',
  marks: 92,
  status: 'Active'
}
{
  _id: ObjectId('6ac5c1b50b19936cd136351d'),
  rollNo: 104,
  name: 'Meera',
  age: 21,
  department: 'ECE',
  marks: 81
}
```

14.db.HARSH.find({ marks: { $not: { $gt: 80 } } })

```
db.HARSH.find({ marks: { $not: { $gt: 80 } } })
{
  _id: ObjectId('6ac5c1b50b19936cd136351b'),
  rollNo: 102,
  name: 'Rahul',
  age: 20,
  department: 'IT',
  marks: 78
}
{
  _id: ObjectId('6ac5c1b50b19936cd136351e'),
  rollNo: 105,
  name: 'Varun',
  age: 23,
  department: 'ME',
  marks: 74
}
{
  _id: ObjectId('6ac5c1b50b19936cd1363520'),
  rollNo: 107,
  name: 'Rakesh',
  age: 21,
  department: 'IT',
  marks: 69
}
```